MILITARY SPECIFICATION

FIRING TRAIN TEST HOUSING, OPERATOR CONTROLLED

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing for Test Housing for use in testing Kill Mechanism and Overlay Assembly.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-45743 - Soldering, Manual Type, High Reliability, Electrical, Electronic, Instrument Communication and Radar for Aerospace and Control Systems, Procedures For
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-M-48245 - Molding Compound, Housing and Plug

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9298406	- Detonating Cord Assembly
9298701	- Housing, Firing Train Test, Operator Controlled
9313637	- Tray for Housing, Timing and and Fuzing - Type 1/Type 2
9313638	- Packing and Marking for Box, Fiberboard for Housing, Timing and Fuzing - Type 1/Type 2

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D-695-(69) - Compressive Properties of Rigid Plastics

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Test housing. - The test housing shall comply with all requirements specified on Drawing (Dwg.) 9298701, all associated drawings and with all requirements specified in applicable specifications.

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3.3 Presence and position of barrier in operation - controlled S&A (OCS&A) assembly. - The OCS&A assembly shall have the barrier present and properly positioned when inspected and shall be marked to indicate that the assembly has passed this inspection.

3.4 Qualification test of piston actuator, detonator and lead cup assembly. - The piston actuator, detonator and lead cup assembly shall function satisfactorily as evidenced by movement of the barrier to the out-of-line position and functioning of the detonatoing cord assembly when temperature conditioned and tested at -25 ± 5°F, +70 ± 5°F and +125 ± 5°F.

3.5 Detonator resistance. - The resistance of the electric detonator shall be not less than 2.9 ohms or greater than 12.0 ohms.

3.6 Piston actuator resistance. - The resistance of the piston actuator shall be not less than 3.9 ohms or greater than 12.0 ohms.

3.7 Continuity of detonator short. - The electric detonator shall be shorted as evidenced by a resistance of not more than 0.2 ohm.

3.8 Continuity of piston actuator short. - The piston actuator shall be shorted as evidenced by a resistance of not more than 0.2 ohm.

3.9 Temperature conditioning and functioning of test housing. - The test housing shall function the detonating cord assembly when temperature conditioned at -25°F ± 5°F, +70 ±5°F, and +125 ± 5°F.

3.10 Ultimate compressive strength of molding compound. - The molding compound used to mold the test housing shall have an ultimate compressive strength of 25,000 psi minimum.

3.11 Void criteria for test housing. - Voids with dimensions larger than 1/8 inch across by 1/16 inch deep are not allowed except within the large cavity below 1/4 inch from the open end where voids with dimensions greater than 1/4 inch across by 1/8 inch deep are not allowed. In addition voids in the S&A and penthouse area of ball cavity are allowed to a maximum dimension of 1/8 inch across by 1/8 inch deep, however, the booster pocket must be intact.

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NOTE: Alternate procedure permits usage of comparison photographs, as approved by the Contracting Officer, to determing accept/reject criteria for housing voids.

3.12 Soldered electrical connections. - Soldered electrical connections on the following assemblies shall comply with MIL-S-45743.

Piston Actuator and Clip Assembly (Dwg. 9298713)
S&A and Laminate Assembly, O.C. (Dwg. 9298703)
Perform Assembly, O.C. (Dwg. 9298702)

3.13 First article inspection. - This specification contains technical provisons for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.14 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. Cleaning methods used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection. -

4.3.1 Submission. - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawings	Quantity
Rod Safety, OCS&A	9298707	25
Barrier, OCS&A	9298708	25
OCS&A	9298706	25
OCS&A and Laminate Assembly	9298703	25
Preform Assembly, O.C.	9298702	25
Housing, Firing Train Test	9298701	25
Housing, OCS&A	9298714	25
Plate, OCS&A	9298715	25
Shield, Detonator	9298463	25
Piston Actuator and Clip Assembly	9298713	25
Molding Compound Test Bars	4.5.7	Two (2) from each batch of molding compound used to mold the testing housing

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-F-48375(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Test Housing and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Barrier, OCS&A (Dwg. 9298708) Examination for Defects	25	-	3.2	4.4.2.2
	Rod, Safety OCS&A (Dwg. 9298707) Examination for Defects	25	-	3.2	4.4.2.1
	Housing, OCS&A (Dwg. 9298483) Examination for Defects	25	-	3.2	4.4.2.3
	Plate, OCS&A (Dwg. 9298715) Examination for Defects	25	-	3.2	4.4.2.4
	Shield, Detonator (Dwg. 9298463) Examination for Defects	25	-	3.2	4.4.2.5
	Piston Actuator and Clip Assembly (Dwg. 9298713) Examination for Defects	25	-	3.2	4.4.2.6

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48375(AR)

PARAGRAPH	TITLE Continuation Test Housing and Components		SHEET 2 OF 2		DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	OCS&A (Dwg. 9298706)				
	Examination for Defects	25	-	3.2	4.4.2.7
	Qualification Test	25	-	3.4	4.5.2
	OCS&A and Laminate Assembly (Dwg. 9298706)				
	Examination for Defects	25	-	3.2	4.4.2.8
	Qualification Test	25	-	3.4	4.5.2
	Preform Assembly, O.C. (Dwg. 9298702)				
	Examination for Defects	25	-	3.2	4.4.2.9
	Housing, Firing Train Test (Dwg. 9298701)				
	Examination for Defects	25	-	3.2	4.4.2.10, 4.4.2.11
	Continuity of detonator and piston actuator shorts	25	-	3.7, 3.8	4.5.5
	Temperature Conditioning and Functioning Test	25	-	3.9	4.5.6
	Ultimate Compressive Strength (All test bars)	-	-	3.10	4.5.7

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of test housings shall contain:

a. Molding Compound (base resin and hardener) from one lot interfix number from one manufacturer.
b. OCS&A Assemblies containing the following:

1. Electric detonators (Dwg. 9245691) from one lot interfix number from one manufacturer (See 4.4.3.4)
2. Lead Cup assemblies (Dwg. 9298456) from one lot interfix number from one manufacturer (See 4.4.3.4).
3. Piston actuators (Dwg. A-66A11301) from one lot interfix number from one manufacturer (See 4.4.3.4)
4. Housing, OCS&A (Dwg. 9298433) from one lot interfix number from one manufacturer
5. Barrier, OCS&A (Dwg. 9298708) from one lot interfix number from one manufacturer.

4.4.2 Examination. -See MIL-A-48078.

a. Sampling Plans. - Unless otherwise specified in in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Rod, Safety, OCS&A	1 OF 1	9298707
			NEXT HIGHER ASSEMBLY: 9298706

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length from top of head to end of thread, min.		0.40%	3.2	Gage
102	Length from bottom of head shank end, min.		0.40%	3.2	Gage
103	Pitch diameter of thread, min.		0.40%	3.2	Gage
104	Major diameter of thread, min.		0.40%	3.2	Gage
105	Diameter of shank end		0.40%	3.2	Gage
106	Defective or missing threads on threaded portion adjacent to shank end		0.40%	3.2	Visual
107	Length from top of head to beginning of thread		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.14	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48375(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298708
4.4.2.2	Barrier, OCS&A				NEXT HIGHER ASSEMBLY 9298706
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness (.150)		0.40%	3.2	Gage
102	Overall height		0.40%	3.2	Gage
103	Height of wider portion		0.40%	3.2	Gage
104	Width of wider portion		0.40%	3.2	Gage
105	Width of narrower portion		0.40%	3.2	Gage
106	Material		-	3.1	Certification
Minor					
201	Width of slot, max.		0.65%	3.2	Gage
202	True position of slot		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.14	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48375(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Housing, OCS&A		9298483
			NEXT HIGHER ASSEMBLY
			9298706

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	See 4.5.8			
Major					
101	Width of barrier locking boss		0.40%	3.2	Gage
102	Height of barrier locking boss		0.40%	3.2	Gage
103	Minor diameter of thread		0.40%	3.2	Gage
104	Pitch diameter of thread		0.40%	3.2	Gage
Minor					
201	Presence of flash, voids, or porosity		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.14	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48375(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Plate, OCS&A				9298715
					NEXT HIGHER ASSEMBLY
					9298706
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Presence of flash, voids, or porosity		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.14	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48375(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298463
4.4.2.5	Shield, Detonator				NEXT HIGHER ASSEMBLY 9298703
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Thickness		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.14	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48375(AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.6	Piston Actuator and Clip Assembly		1	1	9298713
					NEXT HIGHER ASSEMBLY: 9298706
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Soldered connections improper		0.65%	3.12	Visual
202	Evidence of poor workmanship		0.65%	3.14	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	OCS&A		9298706
			NEXT HIGHER ASSEMBLY
			9298703

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Barrier missing or improperly positioned		100%	3.3	Gage/4.5.1
Major	None defined				
Minor					
201	Safety rods loose or not fully seated		0.65%	3.2	Visual
202	Seal washer for safety rod broken or missing		0.65%	3.2	Visual
203	Sealant missing or incomplete		0.65%	3.2	Visual
204	Piston actuator and clip assembly damaged		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.14	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	OCS&A and Laminate Assembly		9298703
			NEXT HIGHER ASSEMBLY: 9298702

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lead cup excessively above flush		0.40%	3.2	Gage
102	Lead cup assembly inverted		0.40%	3.2	Visual
103	Qualification Test				
	a. Low Temperature (-25 ± 5 F)	10-0-1		3.4	4.5.2/4.4.4
	b. Ambient Temperature (+70 ± 5 F)	5-0-1			
	c. High Temperature (+125 ± 5 F)	10-0-1			
Minor					
201	Any component missing or not secured		0.65%	3.2	Visual/Manual
202	Sealant missing		0.65%	3.2	Visual
203	Soldered connections improper		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.14	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298702
4.4.2.9	Perform Assembly , O.C.				NEXT HIGHER ASSEMBLY 9298701
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Detonator or piston actuator leads wrong color or routed to wrong sensor case		0.40%	3.2	Visual
102	Any component or assembly missing or not secured		0.40%	3.2	Visual/Manual
103	"PA" or "DET" marking missing or inconsistent with routing of leads		0.40%	3.2	Visual
Minor					
201	Soldered connections improper		0.65%	3.12	Visual
202	Insufficient wire length or stripped length incorrect		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.14	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Housing Firing Train Test (Prior to Soldering Twisted Wires)		9298701
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Piston actuator resistance	100%		3.6	4.5.4
102	Detonator resistance	100%		3.5	4.5.3
Minor	None defined				

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Housing, Firing Train Test				9298701
					NEXT HIGHER ASSEMBLY
					NONE
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Piston actuator not shorted		100%	3.8	4.5.5
2	Detonator not shorted		100%	3.7	4.5.5
3	Safety rod missing		100%	3.2	Visual
4	Absence of mark indicating presence of barrier		100%	3.3	Visual
Major					
101	Detonator shield missing or loose		100%	3.2	Visual/Manual
102	Diameter of large cavity, min.		0.40%	3.2	Gage
103	Safety rod cannot be turned counterclockwise (free to remove)		0.40%	3.2	Manual
104	Lead cup assembly damaged or coated with matrix material		0.40%	3.2	Visual
105	Functioning Test				
	Low Temperature (-25 + 5 F)	10			
	Ambient Temperature (70 + 10 F)	5	25-1-2	3.9	4.5.6
	High Temperature (125 + 5 F)	10			
Minor					
201	Voids exceeding specified criteria		0.65%	3.11	Visual
202	Evidence of poor workmanship	S4	0.65%	3.14	Visual

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4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starding with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Test methods and procedures. -

4.5.1 Presence and position of barrier of OCS&A assembly. - The OCS&A assembly shall be inspected using a suitable inspection technique that will assure the barrier is present and properly positioned.(e.g. X-ray, mechanical probe, photo optics).

Inspection equipment shall be in accordance with 4.4.4. If any assembly fails to comply with the specified requirement it shall be classed defective and removed from the lot. All assemblies which are accepted by this inspection shall be suitably marked such that the mark will remain visible after the assembly is molded in the test housing. (Non-Destructive Test).

WARNING: The following tests are conducted on assemblies containing an explosive component. Test barricades (where used), procedures and equipment shall have prior approval.

4.5.2 Qualification test of OSC&A and laminate assembly. - The assemblies to be subjected to this test shall be selected from the lot of assemblies to be used in the lot of test housings. The OCS&A assemblies shall contain component lots in accordance with 4.4.1. The assemblies shall have detonating cord assemblies (Dwg. 9298406) suitably attached with the face of the booster axially aligned with and parellel to the face of the lead cup assembly and with a 0.20 inch minimum spacing between the two assebmlies. The assemblies shall be temperature conditioned at the specified temperature for two (2) hours minimum prior to testing. The time between removal from the conditioning equipment until testing shall be 3 minutes maximum. The assembly shall be oriented such that the piston actuator force is applied vertically upward. The piston actuator leads

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and detonator leads shall remain shorted until immediately prior to firing. The OCS&A's shall have successfully passed the radiographic examination specified in 4.5.1. The safety road and the piston actuator short shall be removed. The piston actuator shall be initiated to move the barrier. The detonator short shall then be removed and the detonator initiated. The initiating device for the piston actuator and detonator shall be a 6 volt dry cell or other suitable power source. If any unit fails to initiate at least one inch of detonator cord adjacent to the booster, even if attributed to piston actuator failure, that combination of explosive component lots shall be rejected for use in the lot of test housings. Approved barricades shall be used when assembling and testing these assemblies (Destructive test).

4.5.3 Detonator resistance (prior to soldering twisted lead ends). - The housings shall be tested by untwisting the leads and measuring the resistance using a suitable low current ohmeter with the current limited to 1 milliamp, maximum. Care shall be taken to assure the detonator does not function while performing this test. Observation shall be made for housings which fail to comply with the applicable requirement (Non-Destructive Test).

4.5.4 Piston actuator resistance (prior to soldering twisted lead ends. - The housings shall be tested by untwisting the leads and measuring the resistance using a suitable low current ohmmeter with the current limited to 1 milliamp, maximum. Care shall be taken to assure the piston acutator does not function while performing this test. Observation shall be made for housings which fail to comply with the applicable requirement. (Non-Destructive Test).

4.5.5 Continuity of detonator and piston actuator shorts. - The shorts shall be tested using a suitable low current ohmmeter with the current limited to 1 milliamp maximum. The probes shall be placed on the leads of the explosive component. Observation shall be made for assemblies which exceed the maximum resistance specified in 3.7 and 3.8. (Non-Destructive Test).

4.5.6 Temperature conditioning and functioning of test housing. - Each test housing to be subjected to this test shall have a detonating cord assembly assembled in the cavity with the face of the booster axially aligned with and parallel to

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the face of the booster axially aligned with and parallel to the face of the lead cup assembly and with a 0.20 inch maximum spacing between the two assemblies. The detonating cord assembly may be held in place with tape or other suitable means. Those test housings to be temperature conditioned hot or cold shall be placed in a conditioning box and maintained at the specified temperature for 2 hours minimum. After the conditioning time has elapsed the housing shall be tested at the conditioned temperature and with the arc face resting on a horizontal surface. The safety rod shall be removed. Shorted test cables shall be connected to the piston actuator and detonator leads The test housing shorts (pigtails) shall then be removed by cutting. Using a suitable power source, (e.g. 6 volt dry cell) first the piston actuator shall be fired and then the detonator shall be fired. Test cables shall be color coded or suitably identified to minimize the chance of firing the detonator before the piston actuator. Suitable barricades shall be used when preparing and testing these assemblies. Any unit which fails to fire the piston actuator of the detonator, or fails to function at least one inch of the detonating cord adjacent to the booster shall be classed defective (Destructive Test).

4.5.7 Ultimate compressive strength of each molding compound batch (See 3.10)

4.5.7.1 Batch. - A batch of molding compound is herein defined as the quantity of compound mixed at one time using base resin and hardener from one manufacturer's lot to mold test housings or plugs in one continuous unchanged process. When a continuous-type mixing process is used, a batch is that quantity of molding compound mixed and dispensed in one continuous unchanged process during a (4) hour period or portion thereof using base resin and hardener from one manufacturer's lot.

4.5.7.2 Test bars. - One (1) test bar from each batch of molding compound shall be molded at the same time the batch is used to mold housings. Vibration and evacuation, as necessary may be used to mold the bars. The test bars shall be processed with the items molded from the batch of molding compound which the bars represent, i.e. the bars shall be cured in the same ovens, and subjected to the same temperatures for the same time periods as the items

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they represent. After completion of the cure, the bar shall be tested for ultimate compressive strength using equipment in accordance with 4.4.4. If any specimen fails to comply with the specified requirements, the batch of test housings represented by the test bar shall be rejected. All test housings molded from the same batch as the test bar shall remain segregated and identifiable from other housings until satisfactory completion of these tests.

4.5.7.3 Compressive Strength Test. - The compressive strength of the test bar shall be determined as specified in ASTM D-695 (69) except that the specimen dimensions shall be 0.5 inches by 0.5 inches by 1 inch long or 0.5 diameter by 1 inch long. Tolerances on specimen dimensions shall be ± 0.01 inches. The ends shall be flat and parallel to within .003 inches and shall be perpendicular to the long axis of the specimen. Speed of testing shall be as specified in the ASTM. Results shall be evaluated against the requirement specified in 3.10.

4.5.8 Dimensional control of cast parts. - In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least three (3) parts (as cast) from each cavity shall be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 500 parts. Individual cavity identification must be provided.

If any defective parts are found during qualification of the cavity, the cavity producing the defective parts shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

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4.5.9 Check test for deterioration of piston actuator, electric detonator, and lead cup assembly. - The following test is designed to prevent piston actuators, electric detonators, or lead cup assemblies that may have deteriorated in storage from being assembled into the Test Housing. The test shall be performed on piston actuators, electric detonators and/or lead cup assemblies, as applicable, from each lot which has been stored for more than two years, or stored under adverse conditions for any period of time.

4.5.9.1 Functioning. - Thirty five (35) assemblies shall be selected from each lot and subjected to the qualification test specified in 4.5.2. The thirty five (35) assemblies shall be conditioned and tested as follows:

20 @ -25 ± 5°F

5 @ +70 ± 5°F

10 @ +125 ± 5°F

If any assemblies fail to comply with the specified requirements, including duds, the lot shall be rejected for use in the Test Housing. If any lot is rejected all data generated shall be forwarded to Commander, ARRADCOM, ATTN: DRDAR-QAA-Q, Dover, New Jersey 07801.

5. PACKAGING

5.1 Preservation and packaging. -

5.1.1 Level B. - The test housing shall be preserved and packaged in accordance with Dwg. 9313637.

5.2 Packing. -

5.2.1 Level B. - The test housing, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with Dwg. 9313638 and Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Marking. - Marking of the cartons shall be in accordance with Dwg. 9313638 and Code of Federal Regulations, Title 49, Parts 100-199.

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5.4 Shipping. - When shipment of housings from more than one lot are made at one time, each lot shall be kept separate, the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - The components covered by this specification are intended for use in testing the kill mechanism and overlay assembly.

6.2 Ordering data.- See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. - See MIL-A-48078. Submit equipment designs, as required, to Commander, ATTN: DRDAR-QAA-I, ARRADCOM, Dover, N.J. 07801. Request letter of submittal state contractor, contract number, specification number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Drawings. - Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1345-A071

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